

Name: _____

1. Tom has a income of \$10,000 in the current period, and income of \$12,000 in the future period. Tom does not have to pay any tax. The real interest rate is 0.05 or 5% per period.
 - (a) Write down Tom's budget constraint for current period and future period. Then, derive life-time budget constraint.
 - (b) Tom has the following utility function: $U(c, c') = \ln(c) + 0.95 \ln(c')$. Find Tom's optimal choice for consumption in current and future period. Is Tom a lender or a borrower?
 - (c) Now suppose Tom's current income increase to \$12,000. Find Tom's new optimal choice for consumption in current and future period. Is Tom a lender or a borrower?
 - (d) Draw a graph on the space of (c, c') representing showing the result from part c)
2. Textbook Question: 9.1, 9.8